PRACTICAL EXERCISE

OF

JavaScript

AN ISO 9001:2015 CERTIFIED INSTITUTE

COSMOS

The Real Training Centre

4th Floor, Shiva Blessing – 3, Valentine Circle, Waghawadi Road, Bhavnagar. 8154910092

JavaScript

➢ Introduction to JavaScript

1. Print "Hello, World!" to the console.
2. Print Your Bio Data on screen.

➢ Variables and Data Types

3. Create variables for a name, age, and a boolean for whether you are a student or not.
4. Use typeof to check the data types.

➢ Basic Operations

5. Perform addition, subtraction, multiplication, and division with two numbers.
6. Concatenate two strings and log them.

➢ Conditionals: if, else, switch

7. Create a for loop that prints numbers 1-10.
8. Create a while loop to print numbers 1-5.
9. Use forEach() to log each element of an array.

➢ Arrays: Introduction

10. Create an array of your favorite foods and log each item using a for loop.
11. Add and remove elements from the array using push() and pop().

➢ Array Methods: Advanced

12. Use map() to double each element in an array.
13. Use filter() to return only even numbers from an array.
14. Use reduce() to sum up the elements in an array.

➢ Objects: Introduction

15. Create an object representing a book with properties like `title`, `author`, and `year`.
16. Modify one of the properties and log the updated object.

➢ Objects: Advanced

17. Use object destructuring to extract the title and author from a book object.
18. Create an object with a method that logs the details of a book.

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Functions:

19. Write an anonymous function to add two numbers.
20. Write the same function using an arrow function.

Scope and Closures:

21. Create a function that has a local variable, then define another function inside it that accesses the local variable.

Order Functions:

22. Write a higher-order function that takes another function as an argument and applies it to an array of numbers.

Callbacks:

23. Create a function `greet` that takes a callback to say a personalized greeting.

Promises:

24. Create a promise that resolves after 2 seconds and logs a success message.
25. Handle errors using .catch().

Async/Await:

26. Write an async function that fetches data (simulated with a delay using setTimeout).
27. Use await to handle asynchronous execution.

Let & Const:

28. Compare let and const by declaring variables in different scopes and modifying them.

Destructuring:

29. Use array destructuring to get the first and second elements of an array.
30. Use object destructuring to extract properties from an object.

Template Literals:

31. Create a multi-line string using template literals.
32. Use string interpolation to include variables in a string.

Regular Expressions:

33. Write a regex pattern to check if an email is valid
34. Use regex to extract digits from a string.

➢ Error Handling:

35. Write a function that catches errors when trying to parse invalid JSON.
36. Throw a custom error when a function receives an invalid argument.

➢ The DOM: Introduction:

37. Create a simple HTML page and select an element using JavaScript to change its content.

➢ DOM Events:

38. Add a button to your page and create an event listener to change the text when clicked.

➢ DOM Manipulation: Advanced:

39. Create a dynamic list of items and append them to a webpage.
40. Remove an item from the list when clicked.

➢ Working with JSON:

41. Convert a JavaScript object to a JSON string
42. Parse a JSON string into a JavaScript object.

➢ Classes and Inheritance:

Create a Person class with properties and a method, then create a Student subclass that inherits from Person.

➢ Modules and Imports:

43. Split your code into two files: one with a function and another that imports and calls the function

➢ Local Storage:

44. Store a value in localStorage and retrieve it later.

➢ Event Delegation:

45. Create a list of items and use event delegation to log which item is clicked.

➢ Final Project

46. Build a To-Do list where you can add and delete tasks. Use localStorage to persist tasks

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